Gaiti 2019 CLL — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

The 7 SmartSeq2 plates and 4 patients intermix per-neighbourhood on one continuous, low-cycling manifold (iLISI 1.99 pre-Harmony → 5.53 of 7 samples at θ=5; same-sample-neighbour fraction 0.66→0.21). One connected component, 0 detached islands.

Gate 2 — module geography → GO (non-branching)

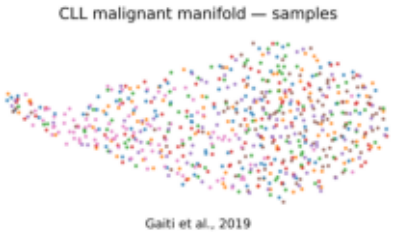
Single (gastrulation+somitogenesis)→neural-crest axis, not a hub with two cross-repressing arms. Gast and pSom co-vary (raw r +0.07, smoothed +0.64) and both oppose Crest (raw −0.57 / −0.59); along pseudotime Gast ρ −0.96 and pSom ρ −0.66 fall while Crest ρ +0.97 rises to a terminal tip. Joint pSom&Crest enrichment 0.43 (<1 → inverse gradient, not a hub).

Gate 3 — cross-repression → NA

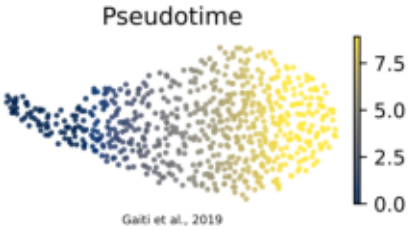
Non-branching topology (principal curve: 2 tips, 0 forks, fold self-approach 3.0, stable across Nodes 15/20/30; a PPT tree finds 0 branch points) — no bifurcation, so the hub cross-repression test does not apply.

STEP FIGURES

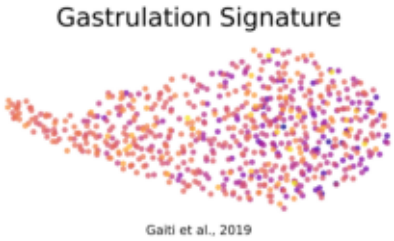
Manifold — samples



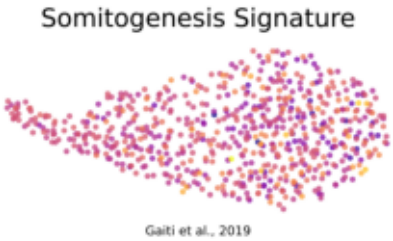
Pseudotime



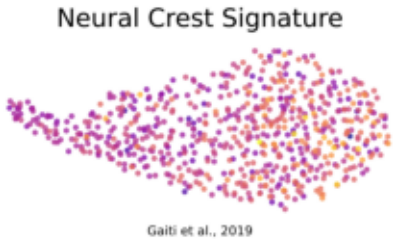
Gastrulation



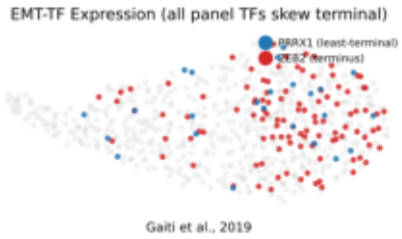
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Chronic lymphocytic leukemia (Gaiti et al., 2019), SmartSeq2 (plate-based) — 672 malignant CD19+ cells, 7 plates / 4 patients. Non-branching Linear-pSom/Crest.

Verdict.
The embryonic pattern reproduces as a non-branching single (gastrulation/somitogenesis)→neural-crest axis (Linear-pSom/Crest) — GO / GO / NA. EMT-TFs ZEB2 (terminus) vs PRRX1 (least terminus-biased); all panel TFs skew terminal, no root-end factor.

Parameter choices

Min-cell threshold	67 (96→48 plate cliff geomean); 7 full plates / 672 cells	**Integration**	Harmony on sample θ=5; n_pcs=20; overdispersed 45 → PCA 44
Geometry / root	principal curve (ElPiGraph), non-branching; root = tip 0 (Gast/pSom-high)	**Hub identity**	NA — non-branching, no convergence hub
EMT-TFs	ZEB2 (terminus) vs PRRX1 (least terminal); no root-end factor; VIM excluded		